

Adversarial Robustification via Text-to-Image Diffusion Models

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01 Data?

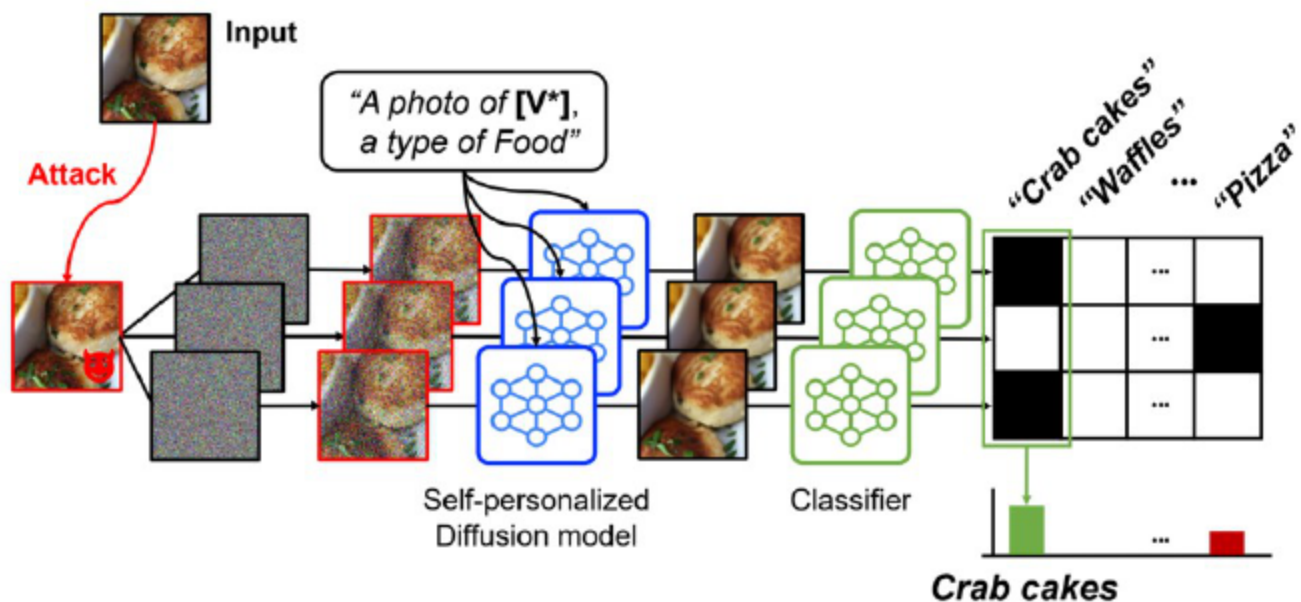
Motivation And Practical Constraints

Off-the-shelf vision models lack robustness and original training data are inaccessible
 Conventional robust training is data-hungry, expensive, and harms clean accuracy
 Need a data-free, scalable, model-agnostic robustification preserving clean accuracy
 Prefer methods with provable guarantees under adversarial perturbations

Limits Of Existing Robustness Transfer

Transfer methods require large external data and are architecture-specific
 Overfit to fine-tuning domain; degrade under domain shift
 Certified smoothing depends on task-trained denoisers, violating data-free goal
 Robustness and clean accuracy often both suffer in practice

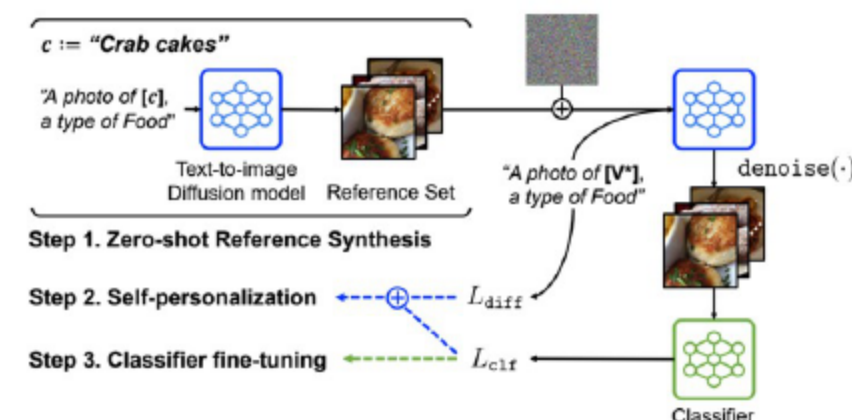
02 Three Main Contributions



- Data-free, model-agnostic robustification via text-to-image denoised smoothing with certificates
- Synthetic references enable diffusion personalization and classifier fine-tuning
- Strong empirical and certified robustness across CLIP and ResNet-50

02 Narrative And Scope

- Formalize data-free objective and denoise-and-classify pipeline
- Introduce timestep correction for super-resolution diffusion
- Design self-adaptation using synthetic references
- Evaluate across zero-shot, ImageNet, and generic classifiers

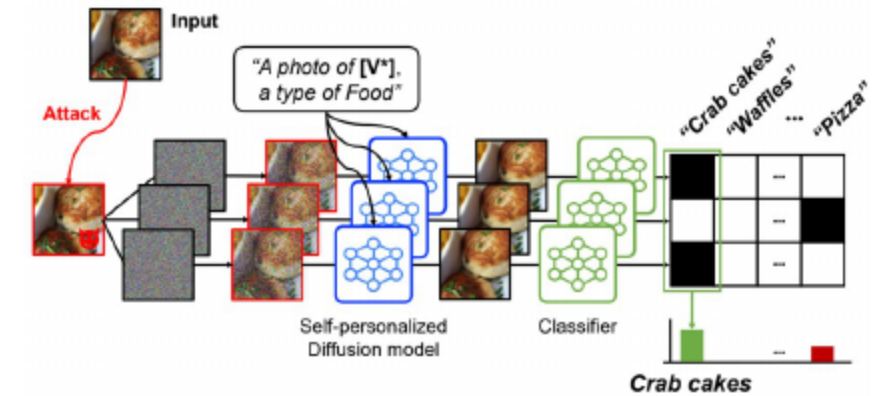


03 Adversarial Robustness And Certification

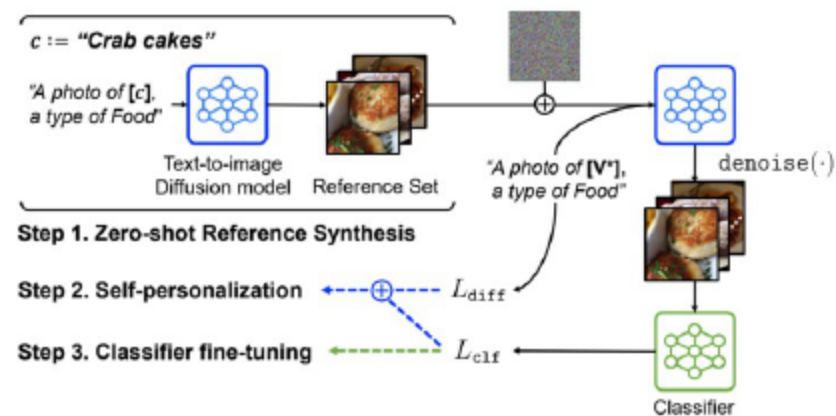
- Robustness: consistent predictions under bounded perturbations
- Randomized smoothing gives probabilistic certificates via noisy voting
- Certified radius hinges on base accuracy under noise

03 Denoised Smoothing Basics

- Denoise noisy input, then classify to form base classifier
- Well-matched denoisers yield strong certified accuracy
- Replace task-trained denoisers with text-to-image super-resolution



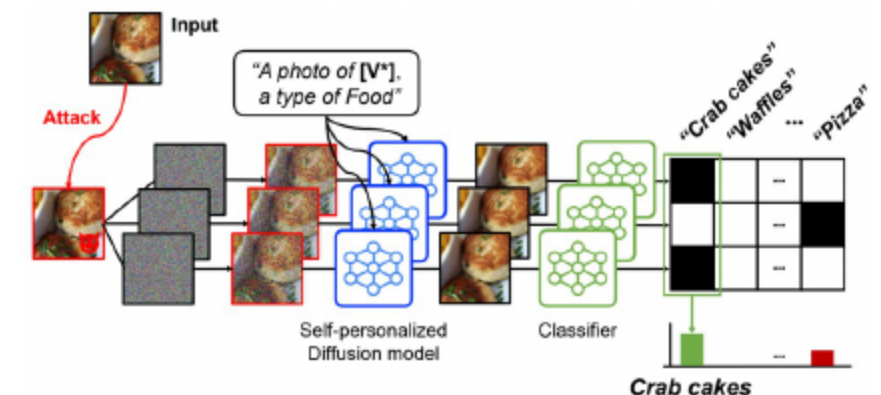
03 Text-To-Image Diffusion Models



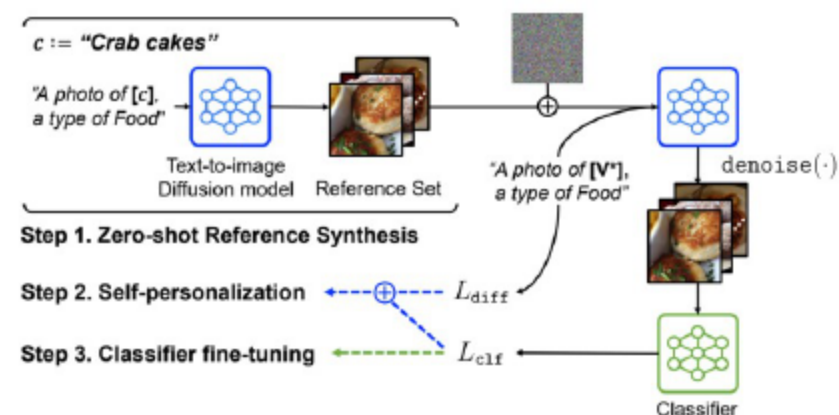
- Cascaded models: text-conditional low-res then super-resolution stages
- Super-resolution modules act as strong pixel-space denoisers
- Compatible with denoised smoothing without task data

04 Pipeline And Zero-Shot Denoising

- Map $x+\delta$ to diffusion timestep via noise schedule
- Use super-resolution denoiser with class text or empty prompts
- Obtain smoothed classifier with certified robustness



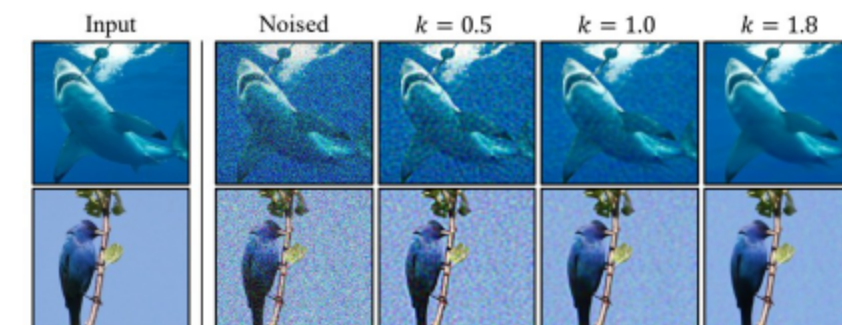
04 Need For Pixel-Space Super-Resolution



- Latent diffusion cannot directly denoise pixel inputs for smoothing
- Pixel-space super-resolution reconstructs content over hallucination
- Condition on label-derived prompts for task relevance

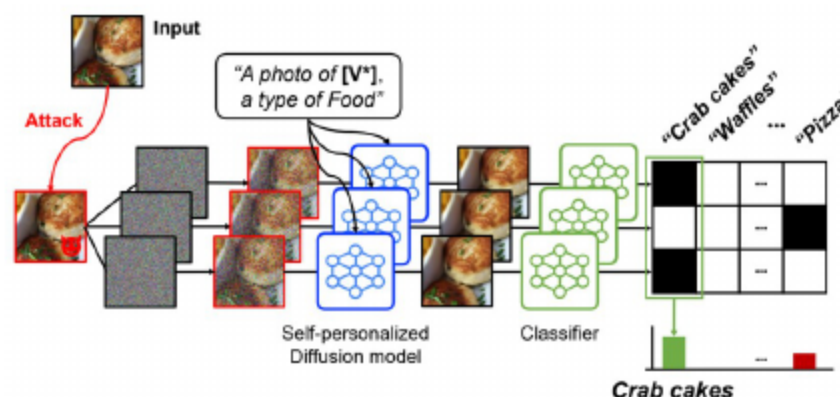
04 Timestep Matching And Correction

- Align noise level to diffusion timestep for all inputs
- Feed scaled noisy input to main and auxiliary branches
- Set higher auxiliary timestep $k > 1$ to suppress high-frequency artifacts
- Improves denoising and certified robustness

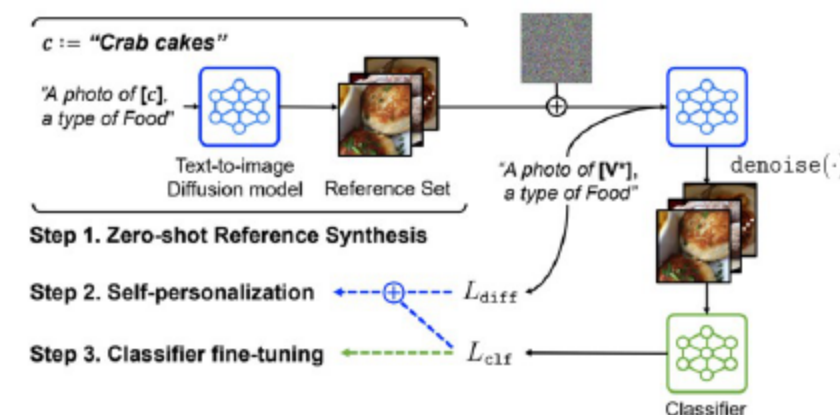


05 Reference Set Synthesis

- Sample small labeled set from text-to-image model using class prompts
- Requires only class names; no real images
- Supports personalization and classifier calibration



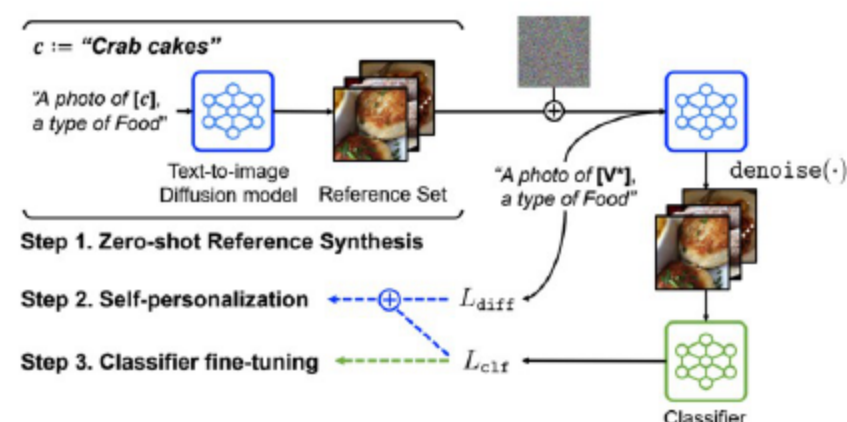
05 Step 1: Classifier-Guided Personalization



- DreamBooth-style fine-tuning with unique identifier prompt
- Classifier-guided loss simulates denoise-and-classify on noisy synthetics
- Freeze text encoder and classifier during personalization

05 Step 2: Classifier Fine-Tuning On Denoised Samples

- Fine-tune classifier on denoised synthetic samples across timesteps
- Denoiser frozen; bridges clean-denoised distribution gap
- Improves robustness while maintaining clean accuracy

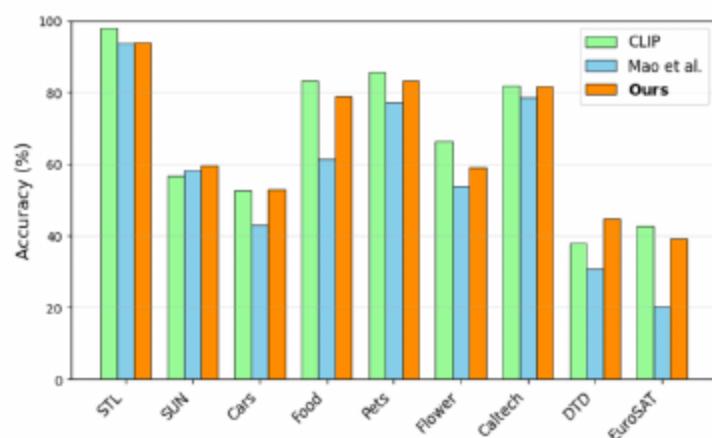


06 Tasks, Models, And Datasets

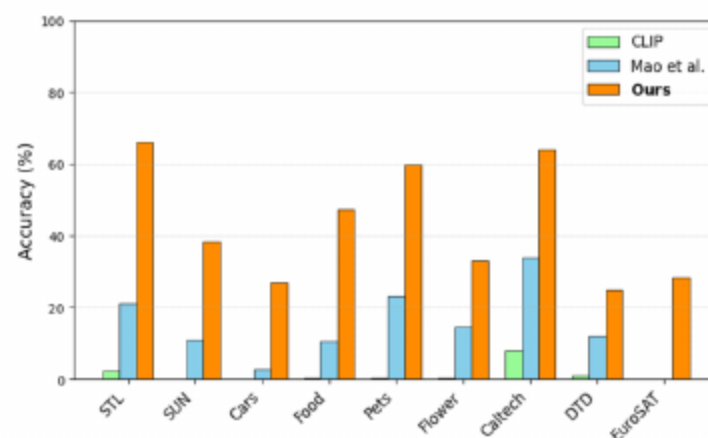
- Robustify CLIP-B/32 across standard and domain-specific zero-shot datasets
- Evaluate on ImageNet; extend to ImageNet-pretrained ResNet-50
- Demonstrate generality beyond CLIP

06 Baselines And Evaluation

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(a) Clean accuracy



(b) Robust accuracy

- Compare against CLIP variants, adversarial training, smoothing baselines
- Report empirical and certified robustness at $\epsilon \in \{0.5, 1.0\}$ and clean accuracy
- Include denoised smoothing with task-trained unconditional diffusion

07 Standard Zero-Shot Benchmarks

	Method	STL	SUN	Cars	Food	Pets	Flower	DTD	Caltech	Average
-	CLIP	97.8	56.8	52.7	83.0	85.7	66.3	37.8	81.9	70.3
$\epsilon = 0.5$	CLIP-Smooth	75.0	46.8	42.1	52.3	66.7	43.5	17.2	68.3	51.5
	Mao et al. [37]	94.8	60.0	48.7	69.7	80.8	57.7	34.0	79.7	65.7
	Ours (Certified)	94.8 (90.4)	58.6 (55.4)	54.1 (49.7)	80.2 (74.5)	83.6 (81.9)	61.4 (58.7)	42.7 (38.9)	81.7 (79.1)	69.6 (66.1)
$\epsilon = 1.0$	CLIP-Smooth	32.4	27.7	40.8	31.3	43.8	36.8	7.0	54.0	34.2
	Mao et al. [37]	93.4	58.2	42.9	61.2	77.0	53.6	30.8	78.5	62.0
	Ours (Certified)	93.8 (80.2)	59.4 (53.6)	52.9 (45.5)	78.8 (64.8)	83.1 (77.7)	58.9 (48.3)	39.1 (32.4)	81.7 (75.7)	68.5 (59.8)

- Substantial empirical and certified robustness gains over CLIP baselines
- Largest improvements at $\epsilon=1.0$; clean accuracy preserved or improved
- Certified often exceeds competitors' empirical robustness

	Method	Robust accuracy (%)			Clean accuracy (%)		
		CropDisease	EuroSAT	ISIC	CropDisease	EuroSAT	ISIC
$\varepsilon = 0.5$	CLIP	0.0	0.0	0.0	20.9	<u>42.6</u>	<u>27.3</u>
	CLIP-Smooth	1.8	<u>13.6</u>	<u>9.0</u>	7.1	16.6	14.3
	Mao et al. [37]	<u>2.2</u>	0.6	4.2	16.0	25.0	26.9
	Ours (Certified)	11.5 (4.3)	29.0 (11.0)	17.6 (5.8)	<u>20.3</u> (16.8)	45.2 (39.0)	35.7 (33.7)
$\varepsilon = 1.0$	CLIP	0.0	0.0	0.0	20.9	<u>42.6</u>	27.3
	CLIP-Smooth	<u>1.2</u>	<u>11.8</u>	<u>4.6</u>	4.3	18.2	10.6
	Mao et al. [37]	0.6	0.0	2.2	<u>16.0</u>	20.4	22.0
	Ours (Certified)	4.9 (0.8)	28.2 (5.0)	8.4 (1.4)	15.8 (8.4)	44.8 (37.8)	<u>26.3</u> (22.0)

- Strong gains on CropDiseases, EuroSAT, and ISIC under domain shift
- Outperforms adversarially fine-tuned CLIP with near-zero robustness
- Maintains or surpasses CLIP clean accuracy

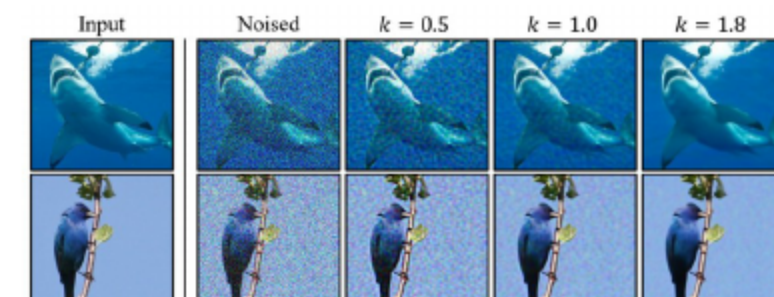
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	Ours (Certified)	4.9 (0.8)	28.2 (5.0)	8.4 (1.4)	15.8 (8.4)	44.8 (37.8)	<u>26.3</u> (22.0)

- Competitive or superior robustness without using real ImageNet data
- Large empirical and certified gains over adversarially fine-tuned CLIP
- Self-adaptation notably boosts certified accuracy

Method	Data-free?	Robust accuracy (%)		Clean accuracy (%)	
		$\varepsilon = 0.5$	$\varepsilon = 1.0$	$\varepsilon = 0.5$	$\varepsilon = 1.0$
Standard Training	✗	5.2	1.0	74.4	74.4
+ Ours (w/o adapt)	✓	<u>56.2</u> (47.0)	44.2 (27.4)	<u>73.0</u> (67.0)	68.8 (57.2)
+ Ours	✓	57.0 (50.4)	47.8 (34.0)	70.4 (68.2)	<u>71.8</u> (60.8)
Adversarial Training [36]	✗	51.0	<u>46.8</u>	55.0	55.0
Randomized Smoothing [13]	✗	55.2 (48.6)	43.8 (37.0)	65.4 (66.8)	55.4 (57.0)
Carlini et al. [7]	✗	<u>56.2</u> (49.2)	45.2 (33.2)	72.6 (67.4)	70.0 (57.8)

- Framework greatly improves ResNet-50 robustness vs. baselines
- Self-adaptation raises empirical and certified robustness further
- Strong clean performance preserved

σ	k	ACR	Certified accuracy at ε (%)					
			0.0	0.25	0.5	0.75	1.0	1.25
0.25	0.5	0.152	36.2	23.0	14.3	6.6		
	1.0	0.229	46.0	35.6	25.2	13.4		
	1.8	0.277	50.8	41.8	29.6	19.2		
0.50	0.5	0.080	13.2	10.2	6.0	4.3	3.4	2.0
	1.0	0.200	28.8	22.0	16.4	11.2	8.4	6.0
	1.8	0.367	42.0	35.4	29.2	23.8	17.6	13.0



- Higher correction factor k yields clearer outputs and higher certified accuracy
- Elevated auxiliary timestep mitigates high-frequency mismatch
- Critical for turning super-resolution into effective denoiser

ϵ	Adapt.		Certified accuracy at ϵ (%)			
	T2I	CLIP	ImageNet	STL	SUN	Food
0.5	X	X	29.6	55.2	28.3	43.6
	✓	X	<u>31.8</u>	66.0	<u>30.7</u>	<u>43.8</u>
	✓	✓	34.2	66.0	32.1	45.7
1.0	X	X	17.6	27.0	17.3	21.8
	✓	X	<u>19.4</u>	<u>40.8</u>	<u>19.3</u>	<u>27.3</u>
	✓	✓	20.6	41.2	22.5	28.9

- Personalization improves certified accuracy across datasets and ϵ
- Classifier fine-tuning on denoised samples adds further gains
- Single adapted pipeline generalizes across multiple σ via randomized timesteps

- Data-free certified robustness via text-to-image super-resolution denoisers
- Self-adaptation preserves or improves clean accuracy and scales broadly
- Future: black-box APIs, new modalities, efficient schedules, zeroth-order adaptation

THANKS!